Open-Loop Thermal Q_{66} Derivation (z-axis)

1. $\sigma^2_{\delta h}$

The variance of the thermal motion per Δt in the wall-normal direction is

$$\sigma^2_{\delta h} \;=\; 4\,k_B T\, a_z(\bar{h}), \qquad a_z(\bar{h}) \;=\; \frac{\Delta t}{\gamma_N\, C_\perp(\bar{h})}, \quad \bar{h} = h/R. \tag{1}$$

2. Chain rule: $\delta h \to \delta a_z$

Starting point. $a_z(h) = \dfrac{\Delta t}{\gamma_N\, C_\perp(\bar{h})}$, with $\bar{h} = h/R$.

Step 1. Treat Δt and γ_N as constants and differentiate w.r.t. $\bar{h}$:

$$a_z \;=\; \frac{\Delta t}{\gamma_N}\cdot\frac{1}{C_\perp(\bar{h})} \quad\Longrightarrow\quad \frac{\partial a_z}{\partial \bar{h}} \;=\; \frac{\Delta t}{\gamma_N}\cdot\frac{\partial}{\partial \bar{h}}\Big(\frac{1}{C_\perp}\Big). \tag{2}$$

Step 2. Apply $\dfrac{d}{dx}(1/u) = -\dfrac{1}{u^2}\dfrac{du}{dx}$ to $1/C_\perp$:

$$\frac{\partial}{\partial \bar{h}}\Big(\frac{1}{C_\perp}\Big) \;=\; -\,\frac{1}{C_\perp^2}\,\frac{\partial C_\perp}{\partial \bar{h}} \quad\Longrightarrow\quad \frac{\partial a_z}{\partial \bar{h}} \;=\; -\,\frac{\Delta t}{\gamma_N}\cdot\frac{1}{C_\perp^2}\,\frac{\partial C_\perp}{\partial \bar{h}}. \tag{3}$$

Step 3.

$$K(\bar{h}) \;\equiv\; \frac{1}{C_\perp(\bar{h})}\,\frac{\partial C_\perp(\bar{h})}{\partial \bar{h}} \quad\Longleftrightarrow\quad \frac{\partial C_\perp}{\partial \bar{h}} \;=\; K\, C_\perp. \tag{4}$$

Substitute back into Step 2:

$$\frac{\partial a_z}{\partial \bar{h}} \;=\; -\,\frac{\Delta t}{\gamma_N}\cdot\frac{1}{C_\perp^2}\cdot K\, C_\perp \;=\; -\,\frac{\Delta t}{\gamma_N}\cdot\frac{K}{C_\perp} \;=\; -\,a_z\, K \qquad \Big[\text{using } a_z = \tfrac{\Delta t}{\gamma_N C_\perp}\Big]. \tag{5}$$

Step 4. Convert from $\bar{h}$ to h via $\partial\bar{h}/\partial h = 1/R$, and take the first-order increment ($\delta a_z = (\partial a_z/\partial h)\,\delta h$):

$$\frac{\partial a_z}{\partial h} \;=\; \frac{1}{R}\,\frac{\partial a_z}{\partial \bar{h}} \;=\; -\,\frac{a_z\, K}{R}, \qquad \delta a_z \;=\; -\,\frac{a_z\, K(\bar{h})}{R}\,\delta h. \tag{6}$$

3. $Q_{66,z}$

$$\boxed{Q_{66,z}(\bar{h}) \;=\; \mathrm{Var}(\delta a_z) \;=\; \Big(\frac{a_z\, K(\bar{h})}{R}\Big)^2\, 4\,k_B T\, a_z(\bar{h})} \tag{7}$$

Implementation note. The two appearances of a_z in (7) refer to different quantities:

- The a_z inside the chain-rule sensitivity $(a_z\, K/R)^2$ is the EKF state estimate $\hat{a}_z$ (slot 6 of the z-axis filter).
- The $a_z(\bar{h})$ inside $\sigma^2_{\delta h}$ is the deterministic function evaluated at $\bar{h}_{zm}$, computed from the position measurement p_m via $\bar{h}_{zm} = (p_m\cdot\hat{w} - p_{\text{wall}})/R$.
- The shape function $K(\bar{h})$ also uses $\bar{h}_{zm}$.

$$R_{22} = \mathrm{Var}\big(a_{xm}[k]\big)$$

1. Starting point

$$\sigma^2_{\delta x_r}[k+1] \;=\; a_{\mathrm{var}} \cdot \delta x_r^2[k] + (1-a_{\mathrm{var}}) \cdot \sigma^2_{\delta x_r}[k] \tag{8}$$

$$a_{xm}[k] \;=\; \frac{\sigma^2_{\delta x_r}[k] - C_n\,\sigma^2_{nx}}{C_d \cdot 4k_BT}, \qquad C_d := 2 + \frac{1}{1-\lambda_c^2}, \quad C_n := \frac{2}{1+\lambda_c} \tag{9}$$

Goal:

$$R_{22} \;:=\; \mathrm{Var}\big(a_{xm}[k]\big) \;=\; \frac{\mathrm{Var}\big(\sigma^2_{\delta x_r}[k]\big)}{(C_d \cdot 4k_BT)^2}.$$

2. Finite-window (white-noise input)

Goal: Show that for an *uncorrelated* (white) input, the filter output variance equals $\dfrac{a_{\mathrm{var}}}{2-a_{\mathrm{var}}} \cdot \mathrm{Var}(X)$.

Notation defined in this section:

- $X[k]$: generic stationary input to the *Exponentially Weighted Moving Average* (EWMA) filter (8) (in § 4 we set $X = \delta x_r^2$).
- $h[k]$: *impulse response* of the EWMA filter — the output sequence when the input is a unit impulse $\delta[k]$.

2.1 Impulse response and convolution form

Solving the recursion (8) with $X[k]$ as input:

$$h[k] \;=\; a_{\mathrm{var}}\,(1-a_{\mathrm{var}})^k, \qquad k = 0, 1, 2, \ldots \tag{10}$$

$$\sigma^2_{\mathrm{out}}[k] \;=\; \sum_{i\ge 0} h[i]\, X[k-i] \quad \text{(convolution form of EWMA).} \tag{11}$$

2.2 Variance for white input

If $X[k]$ is uncorrelated (i.e. $\mathrm{Cov}(X[k], X[k+\tau]) = 0$ for $\tau \neq 0$), summing variances of independent terms:

$$\begin{aligned}
\mathrm{Var}\big(\sigma^2_{\mathrm{out}}[k]\big) \;&=\; \sum_{i\ge 0} h[i]^2 \cdot \mathrm{Var}(X) \\
&= a_{\mathrm{var}}^2\, \mathrm{Var}(X) \sum_{i\ge 0} (1-a_{\mathrm{var}})^{2i} \\
&= \frac{a_{\mathrm{var}}^2}{1-(1-a_{\mathrm{var}})^2}\, \mathrm{Var}(X) \;=\; \frac{a_{\mathrm{var}}}{2-a_{\mathrm{var}}} \cdot \mathrm{Var}(X).
\end{aligned} \tag{12}$$

$$\boxed{\mathrm{Var}\big(\sigma^2_{\mathrm{out}}[k]\big)\,\big|_{\text{white } X} \;=\; \frac{a_{\mathrm{var}}}{2-a_{\mathrm{var}}} \cdot \mathrm{Var}(X)} \tag{13}$$

Reading the factor. $\dfrac{a_{\mathrm{var}}}{2-a_{\mathrm{var}}} \to 0$ as $a_{\mathrm{var}} \to 0$ (long window, much averaging, small variance); $\to 1$ as $a_{\mathrm{var}} \to 1$ (no averaging, output equals input). It is the *noise gain* of the EWMA filter for IID input.

3. Self-correlation inflation (correlated input)

Goal: When $X[k]$ is autocorrelated (consecutive samples not independent), the variance is no longer just (**??**). Derive the multiplicative factor $\mathrm{IF}_{\mathrm{eff}}(s)$ that captures this inflation.

Notation defined in this section:

- $\tau \in \mathbb{Z}$: discrete *lag* between two time indices k and $k+\tau$. $\tau = 0$ is the same instant.
- $C_X[\tau] := \mathrm{Cov}(X[k], X[k+\tau])$: the autocovariance of X. For stationary X, $C_X[\tau] = C_X[-\tau]$.
- $\rho_X(\tau) := C_X[\tau]/C_X[0]$: normalized autocorrelation, $\rho_X(0) = 1$.
- $s := 1 - a_{\mathrm{var}}$: shorthand for the EWMA decay rate.
- $(h \star h)[\tau]$: *self-overlap* of h — the filter's own autocorrelation, computed below.

3.1 Filter self-overlap

Pairs of impulse-response samples separated by lag τ:

$$\begin{aligned}(h \star h)[\tau] \;&:=\; \sum_{k \geq 0} h[k]\, h[k + |\tau|] \\ &= a_{\mathrm{var}}^2\, s^{|\tau|} \sum_{k\geq 0} s^{2k} \;=\; \frac{a_{\mathrm{var}}^2}{1 - s^2}\, s^{|\tau|} \;=\; \frac{a_{\mathrm{var}}}{2 - a_{\mathrm{var}}} \cdot s^{|\tau|}. \end{aligned} \tag{14}$$

At $\tau = 0$, $(h \star h)[0] = a_{\mathrm{var}}/(2 - a_{\mathrm{var}})$, the same factor obtained in (13).

3.2 Variance with correlated input

Generalizing (**??**): when $X[k]$ has nonzero $C_X[\tau]$ for $\tau \neq 0$, every pair of impulse-response samples $h[k]$, $h[k+\tau]$ contributes the cross-covariance:

$$\mathrm{Var}\big(\sigma^2_{\mathrm{out}}[k]\big) \;=\; \sum_{\tau=-\infty}^{\infty} C_X[\tau] \cdot (h \star h)[\tau]. \tag{15}$$

Sanity check. For white X, $C_X[\tau] = 0$ except $\tau = 0$, so only $(h \star h)[0] = a_{\mathrm{var}}/(2 - a_{\mathrm{var}})$ survives in the sum, recovering (13).

3.3 Apply to $X = \delta x_r^2$ — the inflation factor

Substitute (14) in (15) with $X = \delta x_r^2$:

$$\begin{aligned}\mathrm{Var}\big(\sigma^2_{\delta x_r}[k]\big) \;&=\; \sum_{\tau=-\infty}^{\infty} C_{\delta x_r^2}[\tau] \cdot \frac{a_{\mathrm{var}}}{2 - a_{\mathrm{var}}} \cdot s^{|\tau|} \\ &= \frac{a_{\mathrm{var}}}{2 - a_{\mathrm{var}}} \cdot C_{\delta x_r^2}[0] \cdot \Big[\, 1 + 2 \sum_{\tau \geq 1} \rho(\tau)\, s^{\tau} \,\Big], \end{aligned} \tag{16}$$

where in the last line we used $C_X[\tau] = C_X[-\tau]$ to fold negative lags and pulled out $C_X[0] = \mathrm{Var}(X)$. Define:

$$\boxed{\mathrm{IF}_{\mathrm{eff}}(s) \;:=\; 1 + 2 \sum_{\tau \geq 1} \rho(\tau)\, s^{\tau}, \qquad \rho(\tau) := \frac{C_{\delta x_r^2}[\tau]}{C_{\delta x_r^2}[0]}} \tag{17}$$

Reading the factor. $\mathrm{IF}_{\mathrm{eff}}(s)$ is the autocorrelation $i$nflation $f$actor:

- White input: $\rho(\tau) = 0$ for $\tau \neq 0$, so $\mathrm{IF}_{\mathrm{eff}} = 1$ (no inflation, recovering the white-input result).
- Positively correlated input: $\rho(\tau) > 0$ at small lags, giving $\mathrm{IF}_{\mathrm{eff}} > 1$ — autocorrelation reduces the number of effective independent samples seen by the EWMA, so the variance is inflated above the white-noise level.
- The factor s^τ comes from the EWMA's own self-overlap (14): lag-τ correlations are weighted by how much the filter "sees" two samples separated by τ.

4. Combine: R_{22}

Combining § 2 and § 3, the variance of the IIR-estimated random-component power is the baseline factor times the inflation factor times the input variance:

$$\boxed{\mathrm{Var}\big(\sigma^2_{\delta x_r}[k]\big) \;=\; \underbrace{\frac{a_{\mathrm{var}}}{2 - a_{\mathrm{var}}}}_{\text{finite-window (§ 2)}} \cdot \underbrace{\mathrm{IF}_{\mathrm{eff}}(s)}_{\substack{\text{self-correlation}\\ \text{inflation (§ 3)}}} \cdot \mathrm{Var}\big(\delta x_r^2[k]\big)} \tag{18}$$

Through (9) (a_{xm} is a linear function of $\sigma^2_{\delta x_r}$),

$$\boxed{R_{22} \;=\; \mathrm{Var}\big(a_{xm}[k]\big) \;=\; \frac{1}{(C_d \cdot 4k_B T)^2} \cdot \frac{a_{\mathrm{var}}}{2 - a_{\mathrm{var}}} \cdot \mathrm{IF}_{\mathrm{eff}}(s) \cdot \mathrm{Var}\big(\delta x_r^2[k]\big)} \tag{19}$$

with $s = 1 - a_{\mathrm{var}}$, $C_d = 2 + 1/(1 - \lambda_c^2)$, $\rho(\tau)$ and $\mathrm{IF}_{\mathrm{eff}}(s)$ from (17).